PREMKUMAR RAMESH

DESIGN ENGINEER | CAD AUTOMATION ENGINEER

Bengaluru, India | premramesh2001@gmail.com | 9003548094 | www.linkedin.com/in/prem1147ramesh

PORTFOLIO : Premkumar Ramesh - CAD Automation Engineer

SUMMARY

CAD Automation Engineer with 3 years of experience developing CAD customization, API-driven automation, and rule-based engineering solutions using SolidWorks API, AutoLISP, Visual LISP, VBA, and Teamcenter PLM. Experienced in design automation, drawing automation, engineering process optimization, and standards-driven workflows to improve productivity, design consistency, and engineering efficiency.

WORK EXPERIENCE

ACO TECHNOLOGY CENTRE

ASSOCIATE ENGINEER

Project – 1 (SOLIDWORKS) **05/2025 – Present, Bengaluru**

- Developed **SolidWorks API** automation for product configuration and assembly generation, automatically **modifying model parameters**, rebuilding assemblies, and creating customer-specific product variants with new article IDs, eliminating repetitive manual design modifications.

Project – 2 (SOLIDWORKS)

- Developed a **SolidWorks API**-based drawing verification tool that compared released drawings against master references, identified dimension, annotation, and geometry deviations, generated graphical markups, captured comparison screenshots, and exported automated validation reports to Excel, significantly **reducing manual QC effort**.

Project – 3 (AUTOCAD)

- Developed an **AutoLISP**-based drawing automation system for configurable **mesh gratings** that automatically generated section views, top views, dimensions, annotations, layer assignments, and drafting standards from user-defined width, length, material, thickness, and grating-type inputs, reducing manual drafting effort by approximately **90%**.

Project – 4 (AUTOCAD)

- Developed an **AutoLISP** automation tool for **bar grating** generation that performed pitch calculations, U-profile selection, BOM-related note generation, block insertion, dimensioning, and standards-based drawing creation directly from engineering inputs.

GRADUATE ENGINEER TRAINEE

02/2024 - 04/2025, Bengaluru

Project – 1 (EXCEL TO OUTLOOK)

- Developed **VBA-based Excel-Outlook automation** to generate customer-specific effort reports by dynamically refreshing dashboards, capturing report images, generating Outlook draft emails, updating reporting periods, and populating predefined recipient lists, eliminating repetitive reporting activities for engineering teams.

Project – 2 (SOLIDWORKS)

- Developed a **SolidWorks API image-generation** utility that automatically configured export settings, drawing formats, DPI standards, display states, and visualization parameters to generate presentation-ready **PNG outputs** without manual setup.

Project – 3 (AUTOCAD)

- Developed **AutoLISP** utilities for CAD data cleanup, including automated removal of unwanted dimensions generated during SolidWorks-to-AutoCAD conversions, **duplicate block-name correction**, and automated deletion of **obsolete DEFPOINTS layers**, improving drawing quality and standardization.

INTERN

08/2023 - 01/2024, Bengaluru

- Created CAD models and drawings in SolidWorks and Inventor for drainage channels and grating systems under different loading conditions in accordance with **EN1433 and EN 124** standards and performed **GD&T** and **DFMEA** studies.

SKILLS

CAD AUTOMATION: SolidWorks API, AutoLISP, Visual LISP, C#, VBA, .net, NX Open

CAD & PLM: SolidWorks, AutoCAD, Autodesk Inventor, Creo, Catia V5, Teamcenter, Vault Professional, Siemens NX

DESIGNING: GD&T, DFMEA, Engineering Drawings, Product Configuration, Design Validation, Drafting Standards, BOM Management, LCA

WORKFLOW AUTOMATION: Microsoft Excel, Excel VBA, Outlook Automation, Quality Checking Automation

ACADEMIC PROJECTS

Design and Fabrication of IoT-Based Sugarcane Harvesting Bot

- Developed IoT embedded harvesting prototype using SolidWorks for mechanical structure and system integration.
- Developed a control pipeline using **ESP32, ultrasonic sensors, and L293** motor driver for servo/DC motor actuation.

EDUCATION

Bachelor of Engineering

Mechatronics Engineering - Sri Krishna College of Engineering and Technology (9.08 CGPA) **09/2019 - 04/2023, Coimbatore**

ACHIEVEMENTS

- Reduced product configuration and assembly generation by **82%** (50 minutes to 9 minutes) through SolidWorks API automation for APLEX and Concrete Light Shaft products.

LANGUAGES

English (Fluent)**, Tamil** (Native)